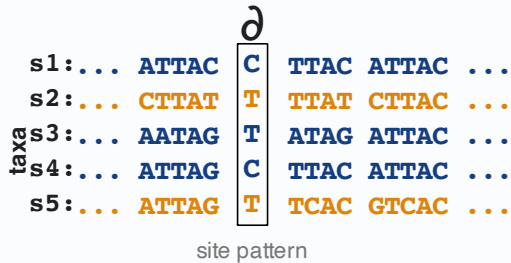


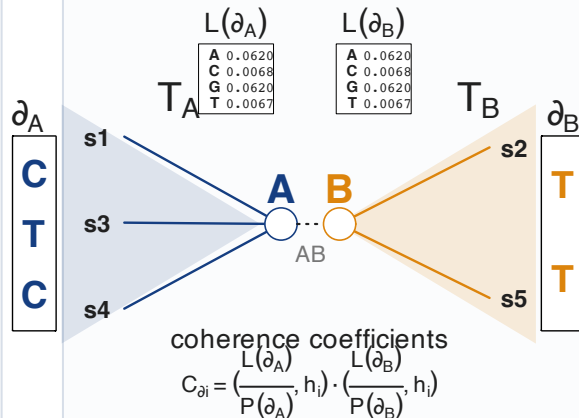
SatuTe in a nutshell

ML-tree inference

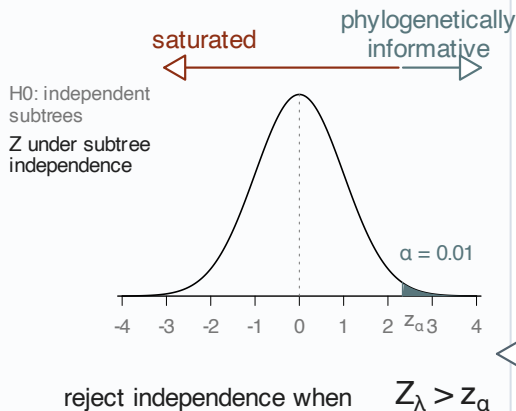
a Alignment and site pattern



b Branch likelihood at AB



d Decision under the null



c Weighted SatuTe statistic

branch-length weights

$$w_i(t) = \exp((\lambda_i - \lambda_1)t)$$

site-pattern coefficient

$$C_{\partial}^{\lambda(t)} = \sum_{i=1}^d w_i(t) C_{\partial i}$$

alignment average

$$\hat{C}_{\lambda} = \sum_{\partial} \frac{n_{\partial}}{n} C_{\partial}^{\lambda(t)}$$

$$Z_{\lambda} = \frac{\sqrt{n} \hat{C}_{\lambda}}{\hat{\sigma}_{\lambda}}$$

$$H_0: \hat{C}_{\lambda} \sim N(0, \hat{\sigma}_{\lambda}^2/n)$$

relative persistence over AB

